Pathway Taken by Water

Exam code: 0610 & 0970

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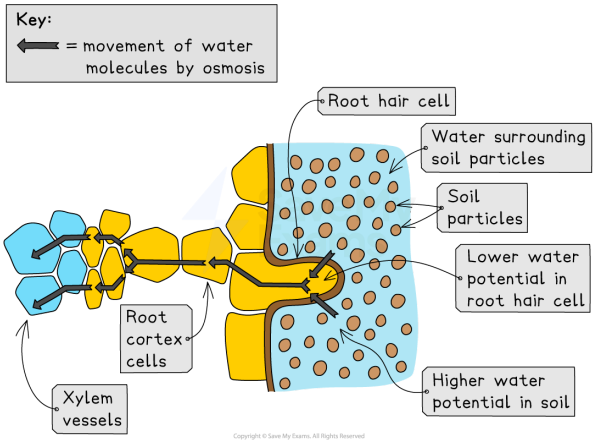
Pathway taken by water

- Water is taken up from the soil into root hair cells
- Water then moves across the root cortex and into the xylem vessels by osmosis



Pathway of water into and across a root

- Water is then transported in the **xylem** to the leaves where it enters the leaf **mesophyll cells**
- So the pathway is:

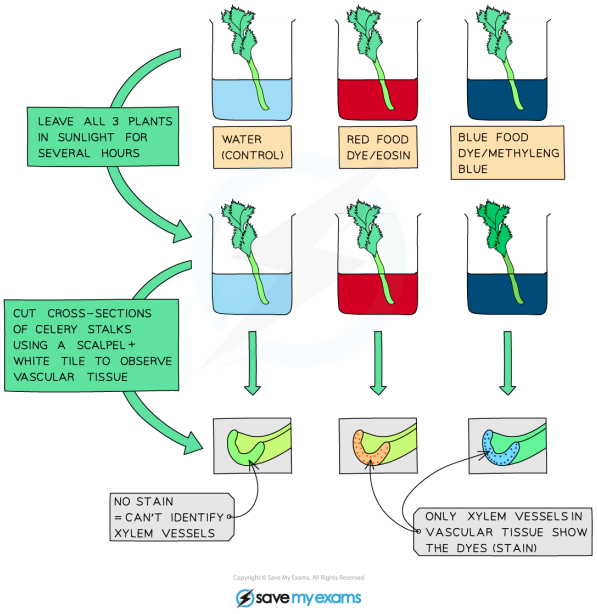
root hair cell → root cortex cells → xylem → leaf mesophyll cells

Investigating water movement in plants

- The pathway taken by water through a plant can be studied using a **suitable stain**, or dye, as follows:
  1. place a plant stem, e.g. celery, into a beaker of water that has had a stain added to it
  2. leave for a few hours, or until the leaves of the celery turn the same colour as the dyed water
     - This indicates that the dye has travelled all the way up the xylem

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it



It is possible to study the pathway taken by water in a plant stem using a suitable stain

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